

1. Objective

The objective of this project is to **predict battery lifetime categories** using only **early-cycle raw measurement data**.

You will reproduce a **battery lifetime classification pipeline**, including:

1. Feature extraction from raw time-series data
2. Derivation of lifetime labels from cycle-of-life information
3. Supervised classification and evaluation

Important:

No features and no labels are given directly.

Both must be **derived by you**. Classify the cells according to their cycle life by labeling the top 45% (longest cycle life) as 'good', the bottom 45% (shortest cycle life) as 'bad', and exclude the middle 10% from the analysis.

2. Dataset Description

2.1 Raw Measurement Data (dataset.xlsx)

- One Excel sheet per battery cell
- Contains **raw time-series measurements**, including:
 - Time
 - Voltage
 - Current
 - Charge / discharge indicators
 - Cycle number

No precomputed features are provided.

You must decide:

- which cycles to use (e.g. early life only)
- which signals are informative
- how to aggregate time-series data into features

2.2 Cycle-of-Life Data (CycleOfLife.xlsx)

For each cell, **battery lifetime** is represented by:

- cell_id
- Cycle Of Life

3. Label Definition

Your Task: Create labels based on the cycle life

Use **distribution-based thresholds**, not fixed cycle numbers. You must:

- Determine the top 45% (longest cycle life) and label them as 'good'
- Determine the bottom 45% (shortest cycle life) and label them as 'bad'
- Exclude the middle 10% from the analysis

4. Feature Extraction

Only **early-cycle data** may be used (e.g. first N cycles).

You must extract features from raw time-series signals. Possible feature categories include (non-exhaustive):

- Voltage statistics (mean, variance, extrema)
- Time spent in voltage ranges
- Capacity-related measures
- Current statistics
- Charge/discharge asymmetry
- Relaxation or rest behavior

Note: There is **no single correct feature set**.

5. Classification Task

- **Input:** feature vectors extracted from early-cycle data
- **Output:** lifetime class label

Think about suitable classifiers. Standard methods include, e.g. k-Nearest Neighbors

6. Evaluation and Analysis

Please track and report the following:

- Train/test split
- Accuracy
- Confusion matrix
- Cross-validation results
- Discussion of:
 - overfitting
 - data leakage risks